Day 19: Odds, odds ratios, and logistic regression

> **Note**
>
> These are reference notes for Day 19. They cover the same material as the Day 19 slides and the self-paced tutorial. Whenever you run into a check-in or a question in the tutorial, write your answer down here.

Where we're going

So far, our response variable has always been **quantitative**. Today the response is **binary** (yes/no, success/failure), so we need a new model: **logistic regression**.

Data: fake news

150 news articles, some real and some fake. Our goal is to explain `type` using features of the article.

- `type`: "real" or "fake"; `type_num`: 0 (real) or 1 (fake)
- `title_words`: number of words in the title
- `title_has_excl`: whether the title has an exclamation point
- `title_caps_percent`: percent of the title in capital letters

A "success" ($Y = 1$) is a _____ article. Always check how your response is coded before modeling!

Q1. Before we do anything else, how do you think `title_has_excl` and `title_caps_percent` are related to `type_num`?

Probability and odds

If π is the probability of an event, the **odds** of the event are

$$\text{odds} = \frac{\pi}{1-\pi} = \frac{P(\text{event})}{P(\text{not event})} \qquad \pi = \frac{\text{odds}}{1+\text{odds}}$$

Probability π	Odds $\pi/(1-\pi)$	How we'd say it
0.5	1	"even odds"
0.75	3	"3 to 1"
0.2	0.25	"1 to 4"

Probabilities are between 0 and 1; odds are between 0 and ∞.

Example: exclamation points

	fake	real	Sum
FALSE	44	88	132
TRUE	16	2	18
Sum	60	90	150

	Title has exclamation	No exclamation
$P(\text{fake})$	$16/18 = 0.889$	_____
$\text{Odds}(\text{fake})$	$16/2 = 8$	_____

Shortcut: from a table, odds = (number of successes) / (number of failures).

The **odds ratio** compares the odds in two groups:

$$OR = \frac{\text{Odds}(\text{fake} \mid \text{exclamation})}{\text{Odds}(\text{fake} \mid \text{no exclamation})} = \frac{8}{_____} = 16$$

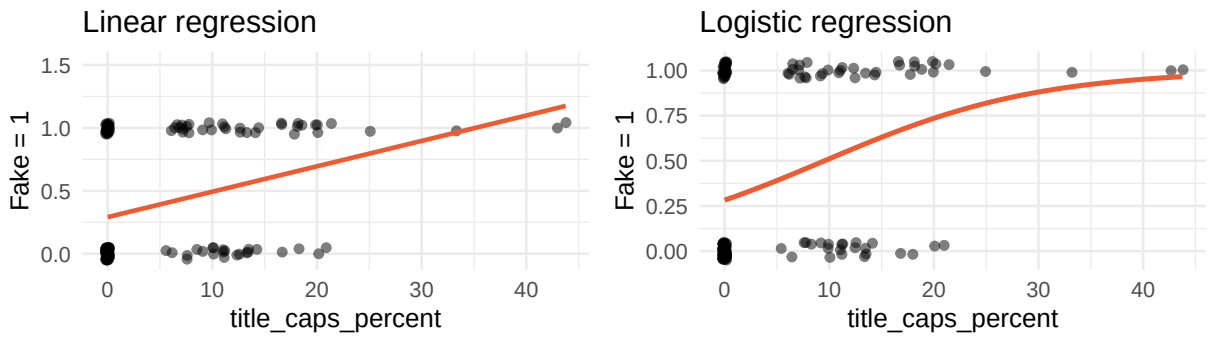
Interpretation: The odds of being fake are 16 times as high for articles with an exclamation point in the title as for articles without one.

> ⚠ Common mistake
>
> An odds ratio is **not** a ratio of probabilities. It is *not* correct to say that articles with an exclamation point are "16 times as likely" to be fake: the ratio of probabilities is only about 2.67.

Q2. An odds ratio of exactly 1 means:

An odds ratio less than 1 means:

Why not linear regression?



A line keeps going forever, so it predicts "probabilities" outside of $[0, 1]$: the linear model predicts 1.17 for the title with the most capital letters (43.75%). Logistic regression gives an S-shaped curve that stays between 0 and 1.

The logistic regression model

$$\log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

where $\pi(x) = P(Y = 1 \mid x)$. The left side is called the **log-odds** or the **logit**. It can be any number, so it's OK for the right side to be a line.

Scale	Formula	Range	Linear in the x's?
Probability	π	0 to 1	_____
Odds	$\pi/(1-\pi)$	0 to ∞	_____
Log-odds	$\log(\pi/(1-\pi))$	$-\infty$ to ∞	_____

Fitting the model in R

```
news_glm <- glm(type_num ~ title_caps_percent, data = news, family = binomial)
```

term	estimate	std. error	t	p-value
Intercept	-0.932	0.220	-4.24	<0.001
title_caps_percent	0.098	0.025	3.84	<0.001

$$\hat{\eta} = \log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = -0.932 + 0.098x$$

where x is `title_caps_percent`.

Interpreting the slope

$\hat{\beta}_1 = 0.098$ is the change in the **log-odds** for a 1-unit increase in x. To get to the odds scale, exponentiate:

$$\widehat{\text{odds}}(x) = e^{\hat{\beta}_0 + \hat{\beta}_1 x} \qquad \Longrightarrow \qquad \frac{\widehat{\text{odds}}(x+1)}{\widehat{\text{odds}}(x)} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 (x+1)}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x}} = e^{\hat{\beta}_1}$$

So $e^{\hat{\beta}_1}$ is an **odds ratio**. Here, $e^{0.098} \approx 1.103$:

> For each additional percentage point of capital letters in the title, the _____ of the article being fake _____ by about 1.10 (an increase of about _____).

For a c-unit increase in x, the odds are multiplied by $e^{c\hat{\beta}_1}$.

What happens to the odds of fake news if the percent of capital letters increases by 10 percentage points?

> The odds _____ by _____

> ⚠ Common mistakes
>
> - $\hat{\beta}_1$ is **not** the change in the probability. In logistic regression, the change in probability for a 1-unit increase in x depends on x.
> - $e^{\hat{\beta}_1}$ **multiplies** the odds; it is not added to them.
> - $e^{\hat{\beta}_1}$ multiplies the **odds**, not the probability.

From log-odds to probability

What's the estimated probability that an article with **20%** capital letters in the title is fake? Work from the inside out:

1. **Log-odds:** $\hat{\eta} = -0.932 + 0.098(20) = 1.023$
2. **Odds:** $\hat{\omega} = e^{\hat{\eta}} = e^{1.023} = 2.78$
3. **Probability:** $\hat{\pi} = \dfrac{\hat{\omega}}{1+\hat{\omega}} = \dfrac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}} = \dfrac{2.78}{1+2.78} = 0.735$

The intercept works the same way: at $x = 0$, the log-odds are _____, the odds are _____, and the probability is _____.

In R, `predict()` does this for you, but by default it gives the **log-odds**. Use `type = "response"` to get probabilities:

```
predict(news_glm, newdata = data.frame(title_caps_percent = 20))
```

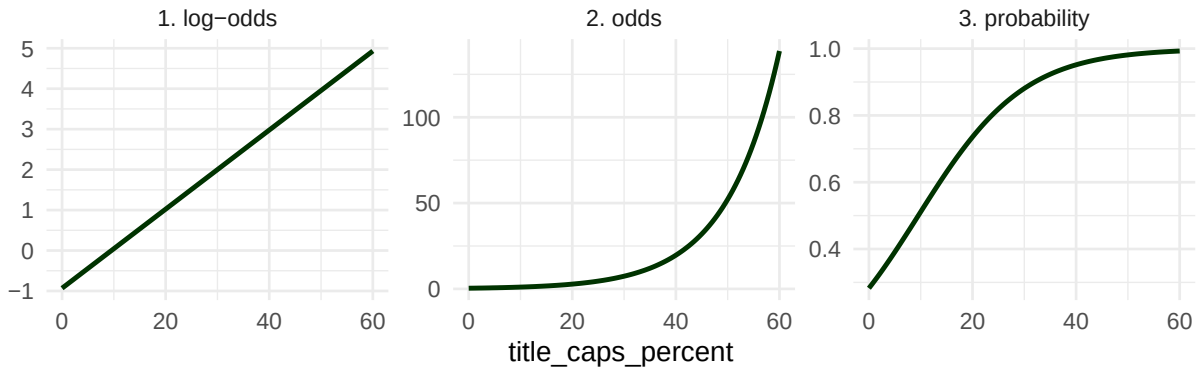
```
       1
1.022656
```

```
predict(news_glm, newdata = data.frame(title_caps_percent = 20), type = "response")
```

```
        1
0.7354896
```

Note: The first line of output (without type = "response") is on which scale? _____

The fitted model on three scales



Q3. Describe the shape of the fitted model on each of the three scales. Why does the probability curve flatten out near 1?

Moving between the three scales

From	To	Formula	In R
log-odds η	odds	e^{η}	`exp(eta)`
odds	probability	$\frac{\text{odds}}{1+\text{odds}}$	`odds / (1 + odds)`
log-odds η	probability	$\frac{e^{\eta}}{1+e^{\eta}}$	`exp(eta) / (1 + exp(eta))`
probability π	odds	$\frac{\pi}{1-\pi}$	`p / (1 - p)`
odds	log-odds	$\log(\text{odds})$	`log(odds)`

A binary predictor gives back the odds ratio

```
excl_glm <- glm(type_num ~ title_has_excl, data = news, family = binomial)
exp(coef(excl_glm))
```

```
      (Intercept) title_has_exclTRUE
              0.5               16.0
```

- $e^{\widehat{\beta}_1} = 16$ is exactly the odds ratio we computed from the table.

Q4. What does $e^{\widehat{\beta}_0} = 0.5$ represent in context?

Your turn: three predictors

`glm(type_num ~ title_has_excl + title_caps_percent + title_words, data = news, family = binomial)`.

Q5. Write out the fitted model on the log-odds scale for articles with and without an exclamation point in the title.

(A) With an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

(B) Without an exclamation point:

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = _____ + _____ \times \texttt{title_caps_percent} + _____ \times \texttt{title_words}$$

Q6. Interpret each slope coefficient in terms of the **odds** of being fake news. (*Note:* templates on last page if you get stuck; but try it on your own first)

`title_has_excl`:

`title_caps_percent`:

`title_words`:

Q7. The estimated probability that an article with an exclamation point, 10% capital letters, and 12 words in the title is fake: _____

Q8. Which predictors appear to be most important for this model? Why?

Q9. What questions do you have about the `summary()` output?

Interpretation templates

Quantity	Interpretation
$\widehat{\beta}_j$	Change in the log-odds of success for a 1-unit increase in x_j, holding other predictors constant
$e^{\widehat{\beta}_j}$, quantitative x_j	For a 1-unit increase in x_j, the odds of success are multiplied by $e^{\widehat{\beta}_j}$, holding other predictors constant
$e^{\widehat{\beta}_j}$, indicator x_j	The odds of success are $e^{\widehat{\beta}_j}$ times as high for the indicated group as for the reference group, holding other predictors constant
$e^{\widehat{\beta}_0}$	The odds of success when all predictors equal 0 (or for the reference group)

R function reference

Function	Purpose
`table(x, y) |> addmargins()`	Two-way table with row and column totals
`glm(y ~ x, data, family = binomial)`	Fit a binary logistic regression model
`coef(model)`, `exp(coef(model))`	Coefficients on the log-odds scale; odds ratios
`summary(model)`	Model summary and Wald tests
`predict(model, newdata)`	Predictions on the log-odds scale
`predict(model, newdata, type = "response")`	Predictions on the probability scale